THE UNIVERSITY OF BRITISH COLUMBIA

CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Layman Summary

The understanding of thermal movement of paricles within a confined space has been studied extensively in order to understand biological, chemical, and physical processes. Although the theoretical understanding of this confined diffusion is well defined in literature, not much work has been done to understnad the effects of electrostatic and hydrodynamic contributions to the size measurements measured from the motion of nanoparticles, as measurements are often instrinsically performed under confinement. This is due to the fact that 3D particle-tracking is not as commonly used as other, higher throughput sizing techniques. By employing developed techniques of nanoparticle tracking in tandem with a full theoretical model, the non-trivial effects of confinement on size-diffusivity were directly investigated and characterized. The work described in the thesis advances the method and tools of 3D nanoparticle analysis both theoretically and in application.

The first section of the thesis describes a theoretical model of Brownian motion based on the confined-Langevin model using a Debye-Huckel electrostatic potential. The model was implemented, based on literature [7], into a computer-based Python simulation in order to parameterize and create prior distributions of particle motion.

The second section describes a modified method of 3D tracking to obtain faster calculations of particle position from optical microscopy images. This was based on an algorithm described in literature [3]. The validation of the method was done on exact point-spread function simulation data with an accuracy of 1 nm.

The third section describes the aquisition of experimental data using a custom interferometric scattering microscope (iSCAT) with image pixel resolution 55 nm aquiring images between 400 and 1000 frames per second. The data consisted of images of scattering fringes from nanoparticle species and was analyzed using the algorithm developed in section 2. The resultant information was accuracte 3D particle tracks. The models developed in sectino 1 were applied to the tracks to obtain infer particle size and charge.

The thesis presents a method of finding particle size, motion, and electrostatics which can present additional tools to analyze nanoparticle characterization, which was the goal of the thesis.

Abstract

Particle confinement allows for extended measurements of suspended biomolecules and nanoparticles using optical microscopy as it keeps it within the field of view. During confinement, the size-diffusivity relationship changes from that in bulk, where the correction factor is typically assumed without consideration of the occurring measurement conditions such as surface charges and the media.

In this work, we show that by combining interferometric scattering (iSCAT) microscopy with microfluidic nanoparticle confinement we can directly measure the positional distribution and depth-dependent diffusivity. We demonstrate how it depends on particle charge and media properties for both reference beads and lipid nanoparticles. Utilizing range-of-depth motion and in-plane diffusivity as complementary diameter estimates, we showcase how 3D particle tracking under confinement allows for nanoparticle diameter quantification in protein-containing media.

The ability to measure the effect of confined diffusivity improves accuracy in trajectory-based particle sizing in varying media, commonly found in cell biology and nanoparticle sizing.[6].

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Contents

1 Introduction **7**

2 Brownian Motion **9**

2.1 Classical Description of Einstein's Brownian Motion 9

2.2 Langevin Dynamics 11

2.3 From Master Equation to Fokker-Planck 12

2.4 Confined Diffusion Processes 13

2.5 Electrostatic Potentials 13

2.6 Equilibrium Positional Distributions for Anomalous Confined Diffusion 15

3 Interferometric Scattering Principles **17**

3.1 Electromagnetic Scattering of of Homogenous Spheres 18

3.2 Scattering Microscopy 20

3.3 Interferometric Light Scattering (iSCAT) 21

3.4 Resolution limits of optical microscopy 22

3.5 Sources of noise in imaging 23

4 Application of confined diffusion to Biological processes **27**

4.1 Diffusion in cells 27

4.2 Brownian motion as a method for measurements of size in biology 28

4.3 Lipid Nanoparticles 28

5 Langevin Simulations **29**

5.1 Simulation Equations 29

5.2 Surface Charge Estimator 30

5.3 Results 31

5.3.1 Mean-Squared Displacement in Confinement 31

5.3.2 Surface Charge estimation 34

5.4 Conclusion 34

6 iSCAT z-estimation Simulations **35**

6.1 Method Overview . . . 35
6.1.1 Image Background Subtraction and Noise Consideration . . . 35
6.1.2 Radial Integral . . . 35
6.1.3 Nuisance parameter Estimation . . . 36
6.1.4 2D Chainlink . . . 36
6.2 Simulated iPSF localization . . . 38
6.2.1 3-parameter Estimation . . . 38
6.2.2 Grid-Search Estimation . . . 39
6.3 Conclusions . . . 39

7 Widefield backscattering iscat for SPT in 3d confinement tracking and analysis 41
7.1 Questions . . . 41
7.2 Reparameterization of Rayleigh scattering of nanoparticles . . . 41
7.3 Fitting model developed in this thesis . . . 41
7.4 Methods . . . 41
7.4.1 Sample Details . . . 41
7.4.2 Optical System . . . 42
7.4.3 Convex-lens induced confinement . . . 42
7.5 Results i guess . . . 42
7.6 Discussion of Results . . . 42

8 Conclusion 43
8.1 Conclusion . . . 43
8.2 Future outlook . . . 43

A Surface Charge Estimator Derivation 47
A.1 $\hat{B}$ MLE . . . 47
A.2 $\mathbb{V}[\hat{B}]$ . . . 48

Chapter 1

Introduction

It is easy to visualize normally sized objects that surround us day-to-day. When speaking about nano-scales, the effects of forces and interactions between nano-objects become many orders larger than those observed in everyday life. In the nano-world, atomic molecules are constantly in motion, exchanging kinetic energy in a chaotic way. The higher the temperature, the more chaos there is - thus, the world is very difficult to describe, predict, and conquer.

Since every chemical reaction in biological and chemical world depends on this motion, it is extremely important to understand the rate of diffusion. This is the reason this subject has been studied by many physcisits, chemists and bioligists.

In the realm of biological processes, there are many such examples involving the particles in the nano size regime. For example, viruses and vesicles are two examples of small molecules which have immense importance in the field. To investigate such systems, it is often the case that to maintain long measurements of the same particle, it is necessary to have some form of confinement. The full understanding of the behavior of these nanoparticles under confinement is thus important, and not complete.

Due to recent advantages of computational power, the tracking of nano-events is possible. Usage of high-precision imaging techniques, high-speed analysis methods can furthen the knowledge about biology that was investigated for the last few hundred years.

The thesis conists of 8 chapters. Chapter 1 is an introduction to the topic of nanoparticles and their relevance in biological and chemical studies. Chapter 2 provides a description of the litearture of confined brownian motion over the last century. Chapter 3 describes the theoretical description of interferometric scattering microscopy (iSCAT) and physical limits that apply in optical theory. Chapter 4 discusses the application of Brownian motion and iSCAT in a biological context, which details regarding lipid-nanoparticles and why they need to be studied on the single-molecule level. Chapter 5 consists of investigations of confined diffusion using Langevin simulations. A simulated system was developed and the size-diffusivity relation in an

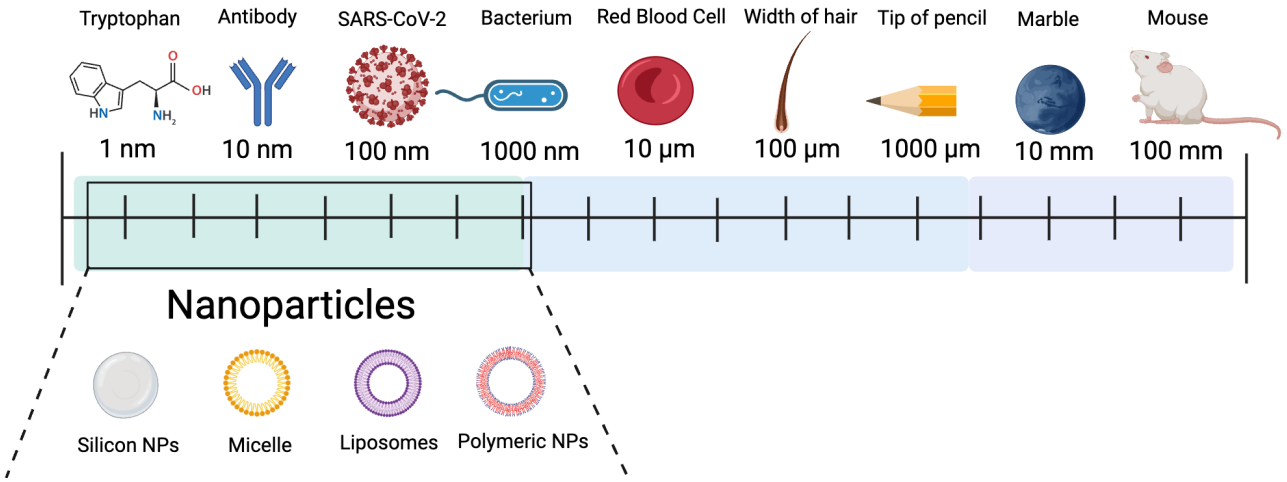


Figure 1.1: Size comparison of various nanoparticles. Nanoparticles are defined as particles where the largest dimension is at most 1000 nm.

electrostaically screened system was investigated. Chapter 6 describes the 3D iSCAT tracking technique developed during this work. The method overview and validation and explicitly developed and validated Chapter 7 describes the method applied to experimental data and utilizes various nanoparticle species, highlighting their varied diffusions. Chapter 8 concludes the world and provides a future outlook on the topic and steps which need to be advanced.

Chapter 2

Brownian Motion

In 1827, Robert Brown observed a stochastic motion of pollen molecules under his microscope [2]. This observation intrigued the likes of many scientists to understand what causes this and how this observation changes the state of understanding of the world. This motion was coined Brownian motion, and has since been applied in many fields.

In the current time, we understand that nanoparticles move around stochastically under the rules of Brownian motion. To be able to measure the particle's characteristics, one can relate their motion and stochasticity to their properties. This chapter will discuss the phenomena of Brownian motion and how it relates to particle properties, environmental ion concentration, viscosity, hydrodynamic boundaries, and confinement.

In this chapter we discuss the process of Brownian motion and Diffusion-type processes. We begin with Einstein's diffusion- equation argument and Langevin's stochastic equation of motion, and then show, through the Fokker-Planck, or Smoluchowski, formalism, that the two are equivalent descriptions of the same process. We next turn to diffusion under confinement, where hydrodynamic wall effects modify the mobility, and to the electrostatic (Debye-Huckel) potentials that govern particle-wall interactions in ionic solution.

2.1 Classical Description of Einstein's Brownian Motion

The physical phenomenon of diffusion can be rigorously described as the emergent behavior atomic particle undergoing random collisions with members of the surrounding environment. In the microscopic limit, where the radius of a particle is sufficiently small and thermal fluctuations dominate its dynamics, this erratic motion is referred to as Brownian motion, or more colloquially, a diffusive process [2].

The classical description of Brownian motion is concerned with how the molecular-kinetic theory of heat relates with microscopic particle suspensions in liquid or gas. In particular, how do

these particles perform movements such that they can be viewed by a microscope? Einstein's argument is one of macroscopic motion. Put simply, one can use a Taylor series expansion over a general concentration function to show a natural property of the system. Using Einstein's original notation, suppose that $\phi(\Delta)$ denotes the probability to find a particle in some space Δ subject to the normalization

$$\int \phi(\Delta) d\Delta = 1 \tag{2.1}$$

Assume that the medium is isotropic such that ϕ is even, $\phi(\Delta) = \phi(-\Delta)$.

Let $f(x,t)$ denote some number of particles at the point x. Einstein takes a finite slice of this volume and equates it to a probability space;

$$f(x, t+\tau) = \int f(x+\Delta, t)\phi(\Delta) d\Delta \tag{2.2}$$

The argument here is that we can taylor expand both sides of this equation,

$$f(x, t+\tau) \approx f(x,t) + \tau \frac{\partial f(x,t)}{\partial t} \tag{2.3}$$

And in space;

$$f(x+\Delta, t) \approx f(x,t) + \Delta \frac{\partial f(x,t)}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.4}$$

Cancelling terms, one finds the classical diffusion equation to nearest order;

$$\frac{\partial f(x,t)}{\partial t} = \underbrace{\left[\frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta\right]}_{D} \frac{\partial^2 f(x,t)}{\partial x^2} \tag{2.5}$$

Where we typically associate the diffusion coefficient with the second moment of the probability distribution

$$D = \frac{1}{\tau} \int \frac{\Delta^2}{2} \phi(\Delta) d\Delta \tag{2.6}$$

The friction coefficient here is related to the solution of the Navier-Stokes equations at a low Reynolds number. Specifically, when there are no intertial forces and $\Re \ll 1$, then the drag force on a particle,

$$\vec{F}_d = 6\pi\eta a v, \tag{2.7}$$

where $\gamma_0 = 6\pi\eta a$, η is the fluid viscosity, and a is the spherical particle's radius. This relation can be directly applied to the form of the diffusion coefficient to recover Einstein's famous forumula;

$$D = \frac{k_b T}{6\pi\eta a}. \tag{2.8}$$

2.2 Langevin Dynamics

In typical experiments, such as that of Robert Brown's pollen molecules, the observed Brownian particle moves around in a fluid volume. If one assumes that this particle exists in the thermydynamic equilibrium, it should obey the equipartion theorem. However, the observed motion appears to have more energy than that of the equipartion at times. This implies something is missing.

This missing component of motion can be described by the bombardment of the surrounding particles around it. In particular, the system experiences a *random* force on it at every time step.

$$m\ddot{\vec{x}} = -\frac{\dot{\vec{x}}}{B} + \vec{F}(t) \tag{2.9}$$

where B is the mobility (drag term is $-\frac{\dot{\vec{x}}}{B}$), which is represented in a viscous friction term, and $\vec{F}(t)$ is a stochastic force term, meaning it takes a random value from some distribution f for every time step t. This model, although easily written, is quite profound and non-trivial to calculate. Here, $x(t)$ is a stochastic process, which has a unique realization for each realization of $F(t)$.

The statistical content of $\vec{F}(t)$ is fixed by the moments and auto-correlation. Notably, this relation is physically associated with the fluctuation-dissipation theory which relates the drag force to the random force. In Brownian motion, this takes the form

$$\langle \vec{F}(t) \rangle = 0, \quad \langle \vec{F}_i(t)\vec{F}_j(t') \rangle = \Gamma \delta_{ij}\delta(t-t') \tag{2.10}$$

Which stands in as delta-correlated noise, or that of a Wiener process. Notably, Γ is constrained by the equipartion theorem. This can be shown by calculating the velocity autocorrelation function (VACF) and choosing the large t limit to agree with the equipartion theorem in 3-dimensions. This fixes the fluctuation constant to being

$$\Gamma = \frac{2k_bT}{B}. \tag{2.11}$$

One can take the ensemble average of this device, and notably, extract correlation information. Rewriting in terms of velocity, and assuming zero drift velocity one finds

$$m\dot{\vec{v}} = -\frac{\vec{v}}{B} + \vec{A}(t), \quad \int_{-\infty}^{\infty} A(t)dt = 0. \tag{2.12}$$

This represents a linear differential equation with a solution hat defines the velocity autocorrelation function. This solution is

$$\langle \vec{v}(0) \cdot \vec{v}(t) \rangle = \frac{3k_bT}{m} e^{-t/(mB)}. \tag{2.13}$$

Furthermore, when one averages the product of the above equation with $\vec{x}$ and integrates over time, we find the following form of the mean-square displacement. Note that $\langle v^2 \rangle = \frac{3k_bT}{m}$, then the solutions of the position mean-squared displacement are written as

$$\begin{aligned} \langle x^2 \rangle &= 6k_bTmB^2 \left[\frac{t}{mB} - (1 - e^{-t/(mB)}) \right] \\ &\begin{cases} \xrightarrow{t \ll mB} \frac{3k_bTt^2}{m} & = \langle v^2 \rangle t^2 \\ \xrightarrow{t \gg mB} \frac{6k_bTmB}{m} t & = (6Bk_bT)t \end{cases} . \end{aligned} \tag{2.14}$$

Letting $D = Bk_bT$, we recover Einstein's relation with ease.

The unique advantage of the Langevin Formulation is that it is readily simulatable [7]. It can be shown that using an Euler-Maruyama scheme, the stochastic differential equation can be integrated over realizations of $F(t)$ using a random number generator, recovering the dynamics of an ideal brownian system [8].

2.3 From Master Equation to Fokker-Planck

To show that the two methods are equal, we can use the typical approach to equilibrium provided by the master equation. Suppose that $f(x,t)dx$ representation that he probability of an arbitrary particle in the ensemble has a displacement between x and $x + dx$. We find that if f is a probability distribution, we can write down the master equation, let W represent the traditional probability transition rate;

$$\frac{\partial f(x,t)}{\partial t} = \int_{-\infty}^{\infty} \left[f(x',t)W(x'|x) - f(x,t)W(x|x') \right] dx' \tag{2.15}$$

This structure is a statement about the probability of positional transitions, which is fundamentally equivalent to the Langevin equation[1] The right hand side can be expanded using the Kramer-Moyal expansion and truncated at second order, yielding the Fokker-Planck Equation.,

$$\frac{\partial f(x,t)}{\partial t} = -\frac{\partial}{\partial x} \left[\sigma(x) f(x,t) \right] + \frac{1}{2} \frac{\partial^2}{\partial x^2} \left[D(x) f(x,t) \right], \tag{2.16}$$

where the diffusive and drift terms are parameterized over the transition probalities,

$$\begin{aligned} \sigma(x) &= \int_{-\infty}^{\infty} \xi W(x|\xi) d\xi \\ D(x) &= \int_{-\infty}^{\infty} \xi^2 W(x|\xi) d\xi. \end{aligned} \tag{2.17}$$

[1] Given a markovian system.

In the case of zero drift, we can clearly see that this form equates to the diffusion equation of the Langevin formulation and the Eisensteinian formulation are the same. This important property allows us to investigate solutions and dynamics of the two systems interchangeably. In particular, relating expected results from simulatory behavior of Langevin problems.

2.4 Confined Diffusion Processes

The effect of diffusion near an interface on a nearby particle is quite particular. The effect on confinement can be first imagined as the effects from the fluid friction flow. Notably, the particle being immersed in fluid invites a stokes friction which effects the perpendicular and parallel flows. Hiding Faxén derived the relations in no-slip boundary conditions near a brownian particle;

$$\begin{aligned}\mu_{\parallel}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1-\frac{9}{16}\left(\frac{a}{h}\right)+\frac{1}{8}\left(\frac{a}{h}\right)^{3}-\frac{45}{256}\left(\frac{a}{h}\right)^{4}-\frac{1}{16}\left(\frac{a}{h}\right)^{5}\right]\\ \mu_{\perp}\left(\frac{a}{h}\right) &\approx \mu_{\infty}\left[1-\frac{9}{8}\left(\frac{a}{h}\right)+\frac{1}{2}\left(\frac{a}{h}\right)^{3}\right]\end{aligned} \tag{2.18}$$

In the case of a two-wall confinement, the form of the diffusion coefficient has a lot of depth in wealth. In application, it has been shown that the form of the Onseen drag in terms of mobility coefficients.

$$\gamma_2(z) \approx \gamma_0 + [\gamma_1(z) - \gamma_0] + [\gamma_1(H-z) - \gamma_0] \tag{2.19}$$

2.5 Electrostatic Potentials

A particle diffusing near a charged wall in an electrolyte will feel a screened electrostatic interaction. The Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds. The potential in a field of ions must be solved in terms of electrostatics in a media. This is described by Poisson's equation

$$\nabla^2\Psi(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{\varepsilon_0\varepsilon_r} \tag{2.20}$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2\Psi(\mathbf{r}) = -4\pi\frac{\rho_j(\mathbf{r})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a cannonical ensemble, which is a good approximation here.

$$\rho_i(\mathbf{r}) = \sum_i C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(r)\right] \tag{2.21}$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2\Psi(\mathbf{r}) = -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \exp\left[-\beta Z_i e_0 \Psi(r)\right] \tag{2.22}$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}
\nabla^2\Psi(\mathbf{r}) &\approx -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} C_i e_0 Z_i \left[1 - \beta Z_i e_0 \Psi(r)\right] \\
&= -\frac{1}{\varepsilon_0\varepsilon_r}\sum_{i=1}^{\infty} \left[C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(r)\right] \\
&= -\sum_{i=1}^{\infty}\left[\frac{C_i e_0 Z_i}{\varepsilon_0\varepsilon_r} - \kappa^2\Psi(r)\right]
\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

When the bulk solution is electroneutral, $\sum_i C_i Z_i = 0$, then the zeroth therm can cleanly drop out, leaving only the approximation;

$$\nabla^2\Psi(\mathbf{r}) = \kappa^2\Psi(\mathbf{r}), \quad \kappa_i^2 = \frac{\beta e_0^2}{\varepsilon_0\varepsilon_r}\sum_i C_i Z_i^2. \tag{2.23}$$

This equation has an obvious solution which are the real exponentials;

$$\Psi(\mathbf{r}) = Ae^{-\kappa\mathbf{r}} + Be^{\kappa\mathbf{r}}. \tag{2.24}$$

The pre-factor A, B is the strength of the electric potential. In the case of ionic potentials, the strength of the prefactor is a number of the number of ions and their charge density surrounding the surface. Thus, one can parameterize this pre-factor based on the geometry.

In the approximations done in this thesis, we assume a linear symmetric potential, as well as a

spherical particle between them. The form of this potential is given by

$$B = |\psi|\Gamma_p\Gamma_w, \tag{2.25}$$

where $|\psi|$ is a fixed amplitude due to charge density, and Γ_w, Γ_p are terms associated with the Stern layers of the wall and the particle;

$$\begin{aligned} |\psi| &= \\ \Gamma_w &= \tanh\left(\frac{e\psi_w}{4k_bT}\right) \\ \Gamma_p &= \tanh\left(\frac{e\zeta_p}{4k_bT}\right) \\ \psi_w &= \frac{2k_bT}{e}\text{asinh}\frac{e\lambda_D\sigma_w}{2\varepsilon_0\varepsilon_r k_bT}. \end{aligned} \tag{2.26}$$

Note that ψ_w is the Stern layer of the wall related to the surface charge density of the wall [5]. This is chosen for a more intuitave measureable of a wall-like strucutre.

2.6 Equilibrium Positional Distributions for Anomalous Confined Diffusion

In the case of interfaces, we have boundary conditions that case the particle to reflect off of the sides. Notably, the probability flux is set to 0. This implies that $\partial_x j = 0$ as well. Then, we need to solve the adjusted Fokker-Plank equation;

$$j = -\frac{\partial}{\partial x}\left[D(x)f(x)\right] - \nabla V(x)f(x) \tag{2.27}$$

Setting j to 0, we find that the solution of this equation defines the equilibrium distribution after a long time.

$$f(x) = \frac{\exp\left[-\beta\int_0^x(\frac{\nabla V(y)}{D(y)})dy\right]}{\exp\left[-\beta\int_{-H}^{H}(\frac{\nabla V(y)}{D(y)})dy\right]} \tag{2.28}$$

This equation defines the equilibrium distribution of the brownian diffusion after a long time.

Chapter 3

Interferometric Scattering Principles

This section is devoted to the description of physical prinicples associated with light scattering techniques. These techniques are widely used in microscopy, as I have used them in measuring where my parteicles are. Scattering measurements are of interest as they allow for label-free imaging with a high signal to noise and high temporal resolution.

We begin with the basics of relating light to electromagnetsm, starting with the Maxwell and Helmholtz relations. We continue discussing the scattering of a various size range of spheres, which is the approximation we take in this thesis. We then discuss the various abberations and Considerations before introducting iSCAT microscopy. We tease the experimental apparatus used in tthis work and compare the method to other methods currently used in the field.

The photon is the fundamental quanta of the electric field as stated in QFT (cite). It has been shown that because of this, the field equations of light obey the fundamental principle that states

$$I = |\vec{E}(\vec{x}, t)|^2, \tag{3.1}$$

where $\vec{E}(\vec{x}, t)$ is defined as an optical field[1]. This principle relation allows us to form models using electromagnetic theory to map out how the optical fields behave.

This chapter will discuss the electromagnetic scattering theory of spheres, discussion of general optical scattering microscopy theory, then discuss the principle of iSCAT and its limitations.

[1]As opposed to an electric field, where $\vec{E}_{\text{optic}}(\vec{x}, t) = \sqrt{\varepsilon_0 n_m c/2}\vec{E}_{\text{elec}}(\vec{x}, t)$. The normalization constant affects how the fields relate to the measured light intensity [4]

3.1 Electromagnetic Scattering of of Homogenous Spheres

To As mentioned, we are tasked with getting the light from the scattered molecules. We approximate them as spheres. Thus, we need to solve the solutions of scattered light from spheres.

We start by using the wave equation for the Electric and Magnetic Fields

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0, \quad \vec{H} = \frac{1}{i\omega\mu_0} \nabla \times \vec{E} \tag{3.2}$$

The solutions of these quantities will be harmonics of the laplacian in the coordinate basis that we choose. For isotropic scattering, the spherical basis is the one of choice. The solutions to this are known as the Vector Spherical Harmonics. Let ψ be a solution to the scalar wave equation

$$\nabla^2 \psi + k^2 \psi = 0 \tag{3.3}$$

, then we can form a generator of the general solution by taking the simplest form of solutions and summing all components. Citing Jackson, we find that the solution to this form is given by the seperable form;

$$\begin{aligned} \psi(r, \theta, \phi) &= \Phi(\phi)\Theta(\theta)R(r) \\ \Phi(\phi) &\in \{\cos(m\phi), \sin(m\phi)\} \\ \Theta(\theta) &= P_n^m(\cos\theta) \\ R(r) &\in \{j_n, y_n, j_n + iy_n, j_n - iy_n\} \end{aligned} \tag{3.4}$$

Where j_n, y_n are the spherical Bessel functions, and their complex additions are the hankel functions. Thus, the vector spherical harmonics are best described by seperating these solutions into even and odd solutions as their prefactor will be cos or sin respectively.

$$\begin{aligned} \vec{M}_{emn} = \vec{\nabla} \times (\vec{r}\psi_{emn}), \quad \vec{M}_{omn} = \vec{\nabla} \times (\vec{r}\psi_{omn}) \\ \vec{N}_{emn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{emn}, \quad \vec{N}_{omn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{omn} \end{aligned} \tag{3.5}$$

These are found in Eqns 4.17 to 4.20 in Bohren.

The incident light can be assumed to be polarized along the x axis without loss of generality. The general form of these equation will then be the sum of each of the spherical vector harmonics (SVH). subject to the energy conservation boundary condition

$$(\vec{E}_i + \vec{E}_s - \vec{E}_1) \times \hat{r} = (\vec{H}_i + \vec{H}_s - \vec{H}_1) \times \hat{r} = 0 \tag{3.6}$$

Citing Bohren, and stating that using orthongality of the SVHs, we have that the exact solution, which is known as the Mie Theory form of scattering off of a sphere, is given below;

$$\begin{aligned}\vec{E_i} &= E_0 e^{ikr\cos\theta}\hat{x} \\ \vec{E_s} &= \sum_{n=1}^{\infty} E_n(ia_n\vec{N^{(1)}}_{e1n} - b_n\vec{M^{(1)}}_{o1n}) \\ \vec{E_1} &= \sum_{n=1}^{\infty} E_n(ic_n\vec{M^{(1)}}_{o1n} - d_n\vec{N^{(1)}}_{e1n})\end{aligned} \tag{3.7}$$

Since we are only interested in the scattered field, the constants that we care about are written below;

$$\begin{aligned}a_n &= \frac{m\Psi(mx)\Psi'(x) - \Psi(x)\Psi'(mx)}{m\Psi(mx)\xi'(x) - \xi(x)\Psi'(mx)} \\ b_n &= \frac{\Psi(mx)\Psi'(x) - m\Psi(x)\Psi'(mx)}{\Psi(mx)\xi'(x) - m\xi(x)\Psi'(mx)}\end{aligned} \tag{3.8}$$

where Ψ, ξ are the Riccatti-Bessel functions.

In the limit of a small particle from far away, we can mae some cancellations in these constants to simplify the work. Notably, the hankel function approches the limit of a spherical wave,

$$h_n(kr) \xrightarrow{kr\gg 1} (-i)^{n+1}\frac{e^{ikr}}{kr} \tag{3.9}$$

In this length-scale, we have that $a_1 \approx \frac{-2i}{3}(ka)^3\frac{m^2-1}{m^2+2}$, and $b_1 \approx 0$. Thus for $n = 1$ we have that $\pi_1 = 1, \tau_1 = \cos\theta$, thus

$$\begin{aligned}M^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \\ N^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right)\end{aligned} \tag{3.10}$$

can be written in the Rayleigh approximation simply and in the form of the full scattering field;

$$\vec{E_s} = \frac{iE_0e^{ikr}}{kr}\left[\hat{\theta}\cos\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\tau_n + b_n\pi_n) + \hat{\phi}\sin\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\pi_n + b_n\tau_n)\right] \tag{3.11}$$

Spoiling the suprise, after plugging in the terms for $n = 1$ we have the following simplification;

$$\begin{aligned}\vec{E}_{s,\perp} &= \vec{E}_{i,\perp}\frac{k^2\alpha e^{ikr}}{r} \\ \vec{E}_{s,\parallel} &= \vec{E}_{i,\parallel}\frac{k^2\alpha e^{ikr}}{r}\cos\theta\end{aligned} \tag{3.12}$$

Then, to find the scattering intensity field, and noting $k = \frac{2\pi}{\lambda/n}$, we have the final scattering field

in the Rayleigh length-scale;

$$\vec{E}_s = \vec{E}_0 e^{ikr} \frac{\sqrt{8}\alpha\pi^2 n_m^2}{\lambda^2 r} \sqrt{1+\cos\theta} \tag{3.13}$$

3.2 Scattering Microscopy

As shown, the description of the scattered light from a sphere is full described by Eqn.3.7, known as the Lorenz-Mie scattering theory [1]. In the limit of particles on the order of the size of the wavelength of light, the Rayleigh model is an easier, equivalent method derived from Mie theory.

To collect the light, there have been three main schemes used in literature; forward, side, and backscattering micrpscopy schemes.

As shown in Fig. 6.1, the scattering distribution taken at different projections widely varies. In addition, the amount of light that gets collected in each of these length-scales also varies.

In the backscattering scheme, the light collected is the sum of the incident light and the scattered light.

$$I(x,y) = |\vec{E}_r + \vec{E}_s|^2 \tag{3.14}$$

given that the incident wave is known as a polarized plane wave, this equation can be rewritten for the scatterd light;

$$I = E_r E_r^* + E_s E_s^* + 2\Re[E_r E_s] \tag{3.15}$$

The incident field will be modulated by a scattering term, which is the Rayleigh scattering amplitude, S, as well as the reference term, R. Thus, the total intensity will be the sum of the scattering amplitude, reference amplitude, and the phase interference between the two fields;

$$I = |E_i|^2 \left[|R|^2 + |S|^2 + 2|R||S|\cos\Delta\varphi\right], \quad \Delta\varphi = \varphi_s\varphi_r \tag{3.16}$$

It should now be noted that the angle of image formation has not been specified. This is of importance as it allows to discuss the various methods of scattering microscopy without loss of generality between methods. Notably, the main difference is the angular content which is obtained from the various scattering models. In the case of spheres, this means using the various angular slices of the resultant scattered field.

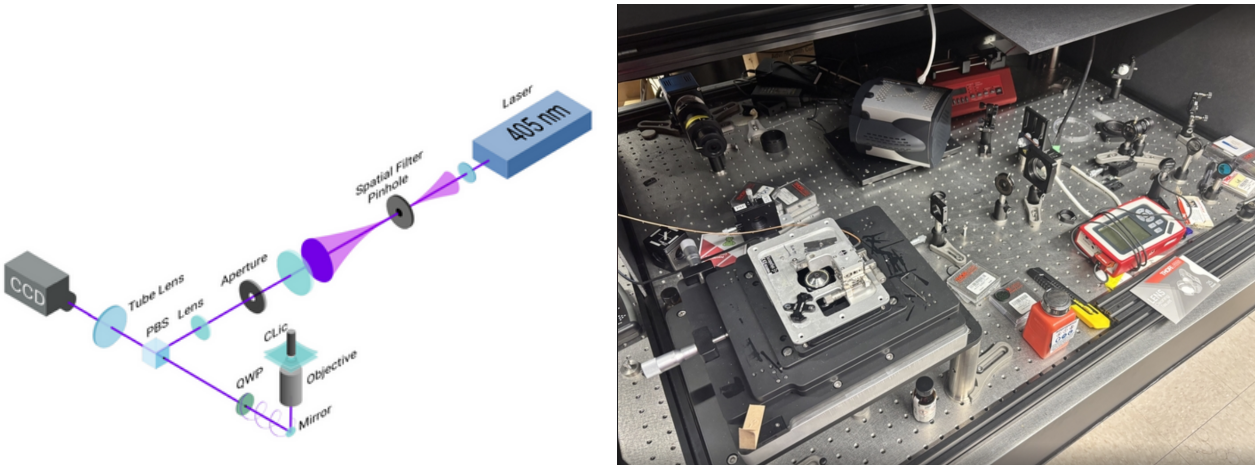


Figure 3.1: Left. iSCAT Schematic. Right. Image of physical system. A 405 nm laser is sent through a spatial filter, passes through an aperture. The beam is sent through a focusing lens which is reflected off a polarizing beam splitter, polarized through a QWP focused on the back plane of the objective. The scatterd light is then polarized again, passing through the polarizing beam splitter and is focused on the CMOS camera. The final image is collected via the CMOS and saved in binary format.

3.3 Interferometric Light Scattering (iSCAT)

The key assumption in iSCAT is the relative magnitude of terms. The reference field is constant. The scattering term $R \approx k^4\alpha^2 = \frac{V^2}{\lambda^4}$. Similarily, the inference term scales as the square root of this. Thus, the scattering term is dominated by the two fields, which we can ignore.

To accurately measure the positions, we can define the contrast metric, which relates the total field to that of a reference field;

$$\begin{aligned} C &= \frac{I - |E_r|^2}{|E_r|^2} \\ &= \frac{|S|^2 + 2|R||S|\cos\Delta\varphi}{|R|^2} \\ &\xrightarrow{|S|\ll|R|} \frac{2|S|}{|R|}\cos\Delta\varphi \end{aligned} \tag{3.17}$$

The phase of the interference consists of a sum of terms consisting of phases accumulated through the traversal of the scattered field. The total phase is dominated by the Gouy phase, which consists of phase accumulated when light is passed through a converging and diverging field, the defocus phase, which can be parameterized to a relative phase shift, as well as the phase accumulated phase difference, which is typical of a Michaelson-Interferomer type system. Experimentally, this system is described in the set of schematics in Fig. 3.1.

3.4 Resolution limits of optical microscopy

The last thing to touch upon is understanding how the image that is being enlarged arrives to the camera, which consists of discussion of optical aberrations.

As mentioned in the formulation for the electric field scatter we have that any optical field consists of a set of plane waves that construct and deconstruct to form a field. That is, the electric field lies in the basis of a fourier series of terms of complex exponentials. This fact allows one to describe the propagation of a wave in throughout z.

$$\vec{E}(x,y,z) = \iint_{\Omega} \vec{E}_0(k_x,k_y) e^{-i\left(xk_x+yk_y+\sqrt{k^2-k_x^2-k_y^2}\right)} dk_x dk_y, \quad \Omega = \{x \in \mathbb{R} : x^2 \leq k^2\} \tag{3.18}$$

We have that the energy of the wave is fixed by the trigonometric relation of the transverse wave vectors. Notably, this is fixed by the relation $|k| = \frac{2\pi}{\lambda}$.

The propagated wave passes through an objective which can be approximated by a lens. Using the ABCD matrix formulation, we can find that to have a clear image, the imaging condition must be satisfied, that is

$$\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{f} \tag{3.19}$$

where z_1, z_2 are the positions of the observer and the object, and f is the focal distance of the lens. Thus, the magnification of the lens is defined as $M = z_2/z_1$. However, to understand the aberrations it is nessecary to have a field formulation of this problem. The simplest way to do this is described using linear optics theory.

Briefly, we define a pupil function $P(x,y)$ which is unitary if $r \leq R$, where R is the radius of lens. The point spread function of the object is then defined as a delta function through a two lens system. This is given by the fourier transform of the point source masked by the pupil function;

$$H(x,y) = \iint_{\Omega} P(x',y') e^{\frac{ik}{2}\left(\frac{1}{z_1}-\frac{1}{f_1}-\frac{1}{f_2}+\frac{1}{z_2}\right)\sqrt{x'^2}+y'^2} e^{-\frac{ik}{z_2}(xx'+yy')} dx'dy' \tag{3.20}$$

Expanding this, one can use a zero-order Hankel transform to integrate over the radius and find that the point-spread function, H, is defined in terms of the NA and the magnification;

$$H(r,z) = \int_0^1 e^{-\frac{ik}{2}\frac{\mathrm{NA}^2}{n}\rho^2 z} J_0\left(\frac{k\mathrm{NA}}{M}\rho r\right)\rho d\rho \tag{3.21}$$

where ρ is the rescaled radial vector $\rho = r'/R$, $\mathrm{NA} = nR/f_1$, and $M = nf_2/f_1$.

In perfect focus, we have that $z = 0$, for $z \neq 0$ we have a defocusing aberration which is defined as the secondary exponential. For a point source, which is diffraction limited, we find that the

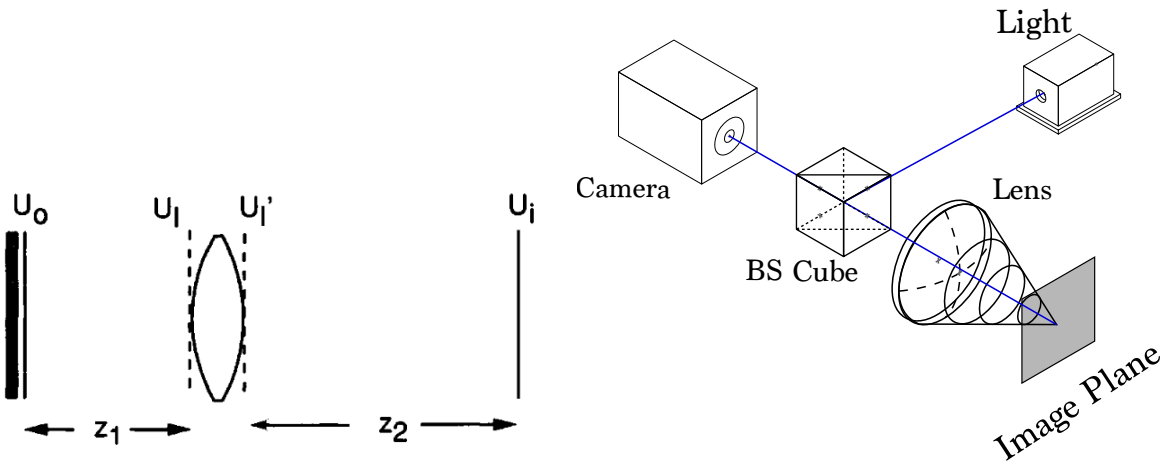


Figure 3.2: Simplest possible imaging setup. Image consists of a plane wave illumination, a lens, and a camera.

following is the limit of the optical microscope;

$$H(r,0) = |\frac{J_1(k\mathrm{NA}r)}{k\mathrm{NA}r}|^2 \tag{3.22}$$

We can solve that the minimum distance between the peak of the PSF and the minumum is solved by $k\mathrm{NA}r = x_0$ where x_0 is the first root of $J_1(x)$. This is classical known as the minimum transverse imaging criterion;

$$r_{\min} = 0.61\frac{\lambda}{\mathrm{NA}} \tag{3.23}$$

Similar results are found for the minimum axial imaging criterion by setting $r = 0$ and integrating.

$$z_{\min} = \frac{2\lambda n}{\mathrm{NA}^2} \tag{3.24}$$

At this point, we invite the fact that not all the light that scatters passes through the lens. Specifically, the range of k_x and k_y are fundamentally limited by the wavelength. The resolution limit is λ/NA. In the case of scattering microscopy, this is not a large problem as the scatteirng pattern is much larger, however, in the limit of two particle near each other, the spatial resolution limit must be enforced.

3.5 Sources of noise in imaging

In imaging, there is a number of sources of abberations that one must account for. To discuss these sources of noise, a list of noises will be formed as one goes through the process of taking an image.

The schematic of an iSCAT microscope was already shown in Fig. 3.1, however, one can use a simplified model system without loss of generality. For a general imaging system, one only

requires a single lens for imaging. This simple setup is shown in Fig 3.2.

The lens will collects light from the object plane, U_o to the imaging plane, U_i, or the camera's plane. The propagation of that wave will collect aberration from the len's point spread function, as well as phase accumulation. The traversal over the point spread fucntion is

$$U_i(x_i, y_i) = \iint_{\mathbb{R}^2} h(x_i, y_i, x_o, y_o) U_o(x_o, y_o) dx_o dy_o, \tag{3.25}$$

where $h(x_i, y_i, x_o, y_o)$ is known as the optical transfer function, or the response of a lens to a point-source, modelled as a delta-function. The len's magnification is incorporated into this using the following approximation;

$$h(x_i, y_i, x_o, y_o) \approx K\delta(x_i \pm Mx_o, y_i \pm My_o) \tag{3.26}$$

where K is a complex constant and M is a real-valued magnification. We can see that because of this, our image is spatially limited by thsi point-spread, which was earlier shown by the diffraction limit.

In addition to the spatial noise, there is a phase accumulation known as the Guoy phase. As the light, which can be assumed as consisting of an eigen-expansion of plane waves, passes through a lens, these plane waves will converge to a gaussian beam with a beam-waist around the focal point. As this focusing occurs, there is a phase that is accumulated known as the Guoy phase. This phase acumulation is equal to

$$\psi(z) = \arctan\left(\frac{z}{z_R}\right), \quad z_r = \frac{\pi w_0^2 n}{\lambda}, \tag{3.27}$$

where w_0 is the waist of the final beam, n is the refractive index, and λ is the wavelength of light. In imaging, as long as the refractive index is constant, there is a total phase shift of π.

In the presence of a background beam, there will be a presence of a background. This background's source is the illumination beam itself, in the case of scattering microscopy. This background noise profile will take the shape of the illumination profile alongside the intensity of the power density.

The camera used will also contain electronic noise associated with it. For CMOS cameras, ther noise present will consist of photon noise from the detectors and readout noise from the data-transfer. These noises can be modelled as Poisson noise from the photon detections and a gaussian for the readout noise.

In the case of using an imperfect lens, or a set of lenses as one has with an objective as one does in high magnification microscopy, there is a risk of spherical aberration. Spherical abberations are caused by imperfections and misalignments in the lens itself. This cases the focal point of the lens to be slighty different depeneding on which area the light is focused by lens. For imaging

purposes, spherical aberrations cause various parts of the image to be out of focus, while others are in focus, creating non-uniform images.

Chapter 4

Application of confined diffusion to Biological processes

As we have described, the diffusivity of a spherical particle is directly linked to a motion that is thermally driven. It happens that in the nature of biological cells, this diffusive motion is a driver for many reactions, such as enzyme-protein coupling, protein-protein coupling, and vesicle binding. Thus, it is of utmost importance to adequaetly understand the confinement effect of diffusion in a given system.

Application-wise, the confinement of nanoparticles is directly related to fabrication and development research. Notably, since nanoparticle fabrication involves methods of quality control, these methods must be able to quantify between different measures of noise impending their measurement. In the case of confinement effects, these effects often hinder diffusion, as described, which can indirectly bias the final reported measurement.

4.1 Diffusion in cells

The cellular system consists of many differfent components co-existing in the cellular cytoplasm. To get a survey of sizes, the organelles are on the order of $1 - 10\,\mu\text{m}$. while smaller proteins and enzymes, such as ribosomes, vesicles, and lysosomes are on the order of $10 - 100\,\text{nm}$.

This beings said, the motion is directly linked to the fluidic properties of the cytoplasm. In the case of a eukaryotic cell, the Reynold's number is low, which implies an overdamped langevin equation. In addition, the thermal scale is that of high temperature, or $\sim 310\,\text{K}$. In addition, the comparison between diffusion and advection is important. For diffusion to dominate we need that the diffusive timescale is must shorter. Here, we find that since the Peclet number is proportional to the characteristics length L, while the reynolds number is L^2. for small characteristic lengths, such as those proposed, we see that the diffusive term is clearly shorter.

4.2 Brownian motion as a method for measurements of size in biology

As previosuly shown, we can infert the stoke's einstein difusion relation to directly measrue the sive of a biological diffusing particle. In particlular, this relation can be explicitly formalized for the hydrodynamic radius of a spherical nanoparticle.

$$a_h = \frac{k_b T}{6\pi\eta D} \tag{4.1}$$

Methods such as nanoparticle-tractiking analysis (NTA) or single-molecule particle tracking (SPT) are based on this method.

However, there are other methods that can be used. As this thesis is not concerned with fluorescent methods, they will not be explained in depth, however, fluorescence correlation spectroscopy (FCS) works on similar ideas of measuring diffusion autocorrelation curves using a fluroescent probe.

In complicment to single molcule, there are bulk methods as well. In particular, dynamic light scattering (DLS) allows to measure the size of moelcules based on their shape function. In particular, the scattering intensity can be weighting to the inverse of a^6, which in turn returns the size of the sample. However, these methods are illposed for polydisperse sample, which is one of the weaknesses of the DLS method.

4.3 Lipid Nanoparticles

The bioligcal molecule of choice for this investigation of confinement is a useful one with lots of applications - the lipid nanoparticle (LNP).

First, it is nessecary to identify with the liposome. The liposome, as a macroscopic object, is a spherical membrane of lipid subunit, each carrying a charge and hydrophobicity. The size of a single submit is on the level of angstroms, however, the final liposome can be between $1\,\text{nm} - 10\,\mu\text{m}$. Liposomes and cellular vesicles are very similar, which naturally posed the question of drug delivery using a liposome model.

In recent years, the LNP was formulated - a collection of lipid based walls that are used to encapsulate a drug or cargo. These have applications in drug delivery and have been using in the Sars-Covid vaccine in 2020. These molecules are very useful and thus it is important to know how to improve them.

Modern tools to image and characterize the LNP are split into single-molecule techniques and bulk assays. In the realm of single-molecule methods, CLiC has been shown to be a highthroughput, accurate, and realiable method to charaterise these particle's properties.

Chapter 5

Langevin Simulations

To illustrate the wealth of information and form prior distributions for the confined diffusion model, we can simulate the diffusion process of a control molecule diffusing in a confined electrostatic environemnt.

The goal of this project was to model the diffusion process of confined diffusion using literature, and utilize pre-existing theory to for a statistical estimator of surface charge from diffusion dyanmics.

5.1 Simulation Equations

The simulation scheme follows that descirbed in [7]. A brownian particle diffuses between two walls such that (x, y) are perpendicular to the walls, and z is parallel. The electrostatic force that the particle feels is then only in the z direction. The diffusivity of the particle will depend on the distance of the walls. The langevin equation simulated,

$$m\ddot{\vec{x}}(t) = -\lambda\dot{\vec{x}}(t) + \vec{F}(\vec{x}(t)) + \sqrt{2k_bT\lambda}\vec{W}(t), \tag{5.1}$$

is first seperated into transverse and axial components, where F is the electrostatic force, λ is the friction. Here, we can re-write the diffusion coefficient using the Stokes-Einstein relation,

$$D(z) = \frac{k_bT}{\lambda(z)}. \tag{5.2}$$

Then, in overdamped regime, where $m\ddot{\vec{x}} \ll 0$, we can write the equations of motion as

$$\begin{aligned} \frac{dx}{dt} &= \sqrt{2D_x(z)}W_x(t) \\ \frac{dz}{dt} &= \sqrt{2D_z(z)}W_z(t) + \frac{F_z(z)}{\lambda(z)}, \end{aligned} \tag{5.3}$$

as descirbed in Eqn. 7 of [7]. This is exceptionally valid for particles between $10\,\text{nm} - 100\,\mu\text{m}$ as this is exactly the overdamped regime for many particle materials, such as PMMA or coated polystyrene.

To simulate this equation, we use an Euler-Maruyama scheme. To illustrate this, we can rewrite the above equations in the Ito form, replacing lowercase variables with their captial, stochastic, equivalents:

$$\begin{aligned} dX_t &= \sqrt{2D_x(z)}dW_x \\ dZ_t &= \sqrt{2D_z(z)}dW_z + \frac{F_z(z)}{\lambda(z)}dt, \end{aligned} \tag{5.4}$$

The simulated system of equations is then the pair of equations;

$$\begin{aligned} x_{n+1} &= x_n + \sqrt{2D_x(z_n)}W_{x,n}\Delta t \\ z_{n+1} &= z_n + \left[D_z'(z_n) - \beta D_z(z_n)V'(z_n) + \sqrt{2D_z(z_n)}W_{x,n}\right]\Delta t \end{aligned} \tag{5.5}$$

where $V(z)$ is the electrostatic potential, $\beta = 1/k_bT$, and Δt is the discretized time.

5.2 Surface Charge Estimator

As described earlier, the relations for the effective potential are described if one knows the particle properties and surface charge. In terms of measureables, electrostatic interactions can be fully described by the following equation,

$$U(z) = B\left[e^{-(H-z)/\lambda_D} - e^{-(H+z)/\lambda_D}\right], \tag{5.6}$$

where the prefactor is described by the map $f : (\zeta_p, \sigma_w, a, \lambda_D, T)$,

$$B = C \cdot \tanh\left(\frac{e_0\zeta_p}{4k_bT}\right) \cdot \tanh\left(\frac{e_0\psi_w}{4k_bT}\right) \tag{5.7}$$

$$C = \frac{64\pi\varepsilon_0\varepsilon_r a k_b T}{e_0^2} \tag{5.8}$$

$$\psi_w = \frac{2k_bT}{e_0}\text{asinh}\left[\frac{\sigma_w e_0 \lambda_D}{2\varepsilon_0\varepsilon_r k_b T}\right] \tag{5.9}$$

where ζ_p is the zeta-potential of the particle, ψ_w is the stern-layer potenital for the surface walls, a is the particle radius, e_0 is the elementary charge value, and λ_D is the debye-length.

The surface charge of the walls can be estimated if one knows the electrostatic amplitude potential. Since the electrostatic force is a drift term, one can construct a maximum likelihood estimator as a standard result from the SDE. This estimator is derived in App. A.

$$\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z}\left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}} \quad , \mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)}. \tag{5.10}$$

If the amplitude of the electrostatic interactions is known, one can invert the amplitude using relations in Eqn. 5.7. This inversion requires at least the zeta-potential of the particle, the size, and debye-length.

5.3 Results

We will simulate the priors of a Polystyrene bead diffusing with neutral charged walls, as well as those with varied levels of screening. In turn, we will be able to showcase the differnt cases of the known confined diffusion theory, as well as establishing prior curves for this work. In addition, the simulated curves will be compared to an effective theoretical model. For a given simulation, we can generate a realization of the dynamics following the aforementioned SDEs. These realizations approach the boltzmann limit as $t \gg \tau$. Some such realizations are shown in Fig. 5.1

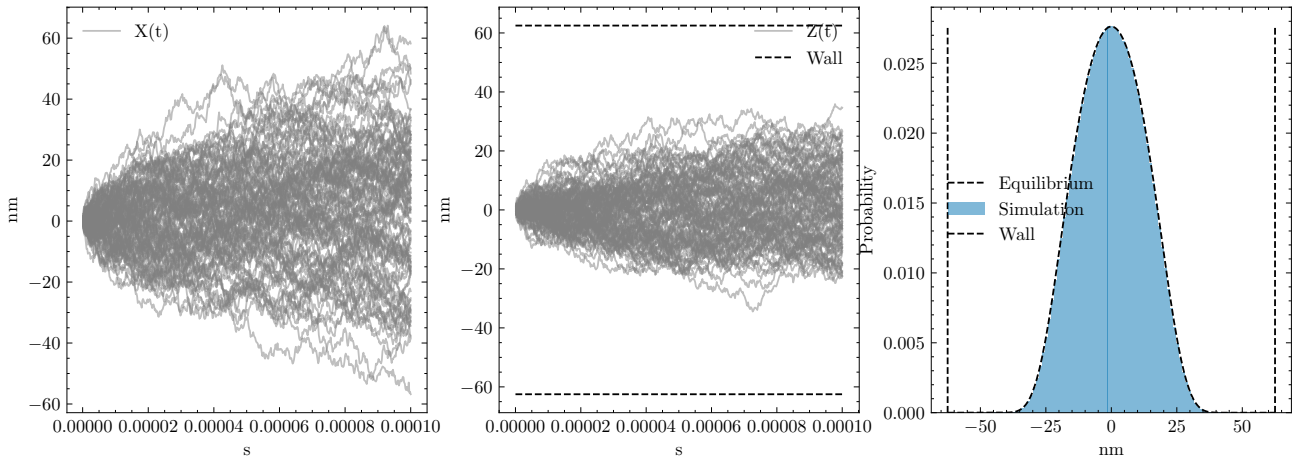


Figure 5.1: Confined Brownian Particle realization of z diffusion in electrostatic potential. $a = 50\,\mathrm{nm}, \zeta_p = -30\,\mathrm{mV}, \sigma_w = -2\,\mathrm{mC/m^2}, L = 225\,\mathrm{nm}, \lambda_D = 10\,\mathrm{nm}, T = 300\,\mathrm{K}, \eta = 1\,\mathrm{mPa\,s}$. Left. Particle trajectory vs time for 100 realizations. Right. Histogram of steps vs. Boltzmann distribution.

5.3.1 Mean-Squared Displacement in Confinement

For a given experiment, one aims to measure something from knowing the trajectory of the particle. The simplest is an estimate of the size of the particle. One can do this using the transverse and axial trajectories obtained from the Langevin equation as a method to test the variance of the estimator given a known ground truth. In the case of an eletrostatic potential, we can directly how measurements which do not account for screening effects change the measurment.

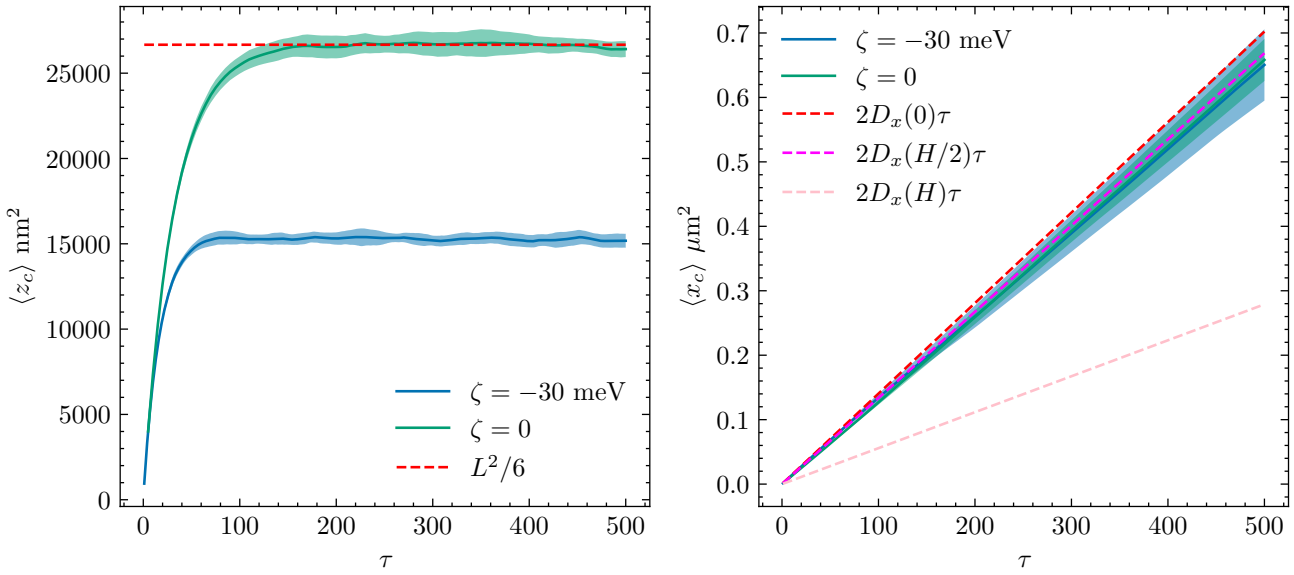


Figure 5.2: Comparison between MSd curves of two highly differnet confined diffusors. Exposure time $\tau = 200\,\mu s$, $\lambda_D = 10\,nm$.

The notation of a higher confinement with respect to the zeta-potenital is intrinsically linked to the electrostatic potential that the particle feels as it explores the well. This relationship between the effective confinement length can be solved by solving the Green's function for the FP-equation in the presence of an electrostatic potential. However, this happens to be non-analytical in the case of the Poisson-Boltzmann potenital approximated in the Debye-Huckel theory. However, using the developed simulation class, this relationship can be directly measured. The same settings were used while fixing the zeta-potential of the particle, and varying the ionic concentration of the solution, which is inversely proportional to the debye-length,

$$\lambda_D \propto \frac{1}{\sqrt{C}}, \tag{5.11}$$

where C is the ionic concentration of a dilute eletrolyte compound [CITE]. To model this, the simulation settings are chosen as such $a = 50\,nm, L = 500\,nm, T = 300\,K, \Delta t_i = 0.2\,\mu s, \tau = 200\,\mu s, \eta = 1\,mPa\,s$ and the debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyene bead, fixed at $\sigma_w = -2\,mC/m^2, \zeta = -30\,mV$. The relationship between the effective confinement length and the ionic concentration is shown in Fig. 5.3

The relationship between the effective confinement length, as well as the relationship between relative diffusion and concentration of ions begs the question of if this can be used as a particle filter to distinguish classes of particles. Here we investigate that question regarding the amount of confinement length compared to the zeta-potential of a particle at various debye-lengths. This relationship is described in Fig. 5.5.

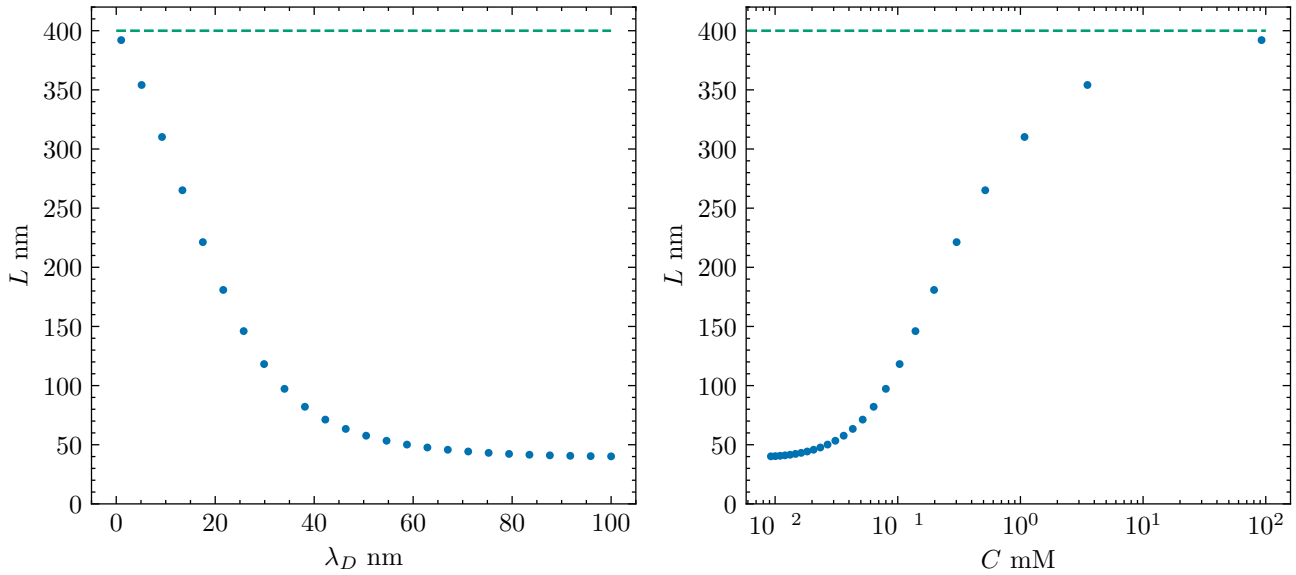


Figure 5.3: Effective Confinement length plotted against the solution's debye length. $a = 50\,\mathrm{nm}, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$ and the debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyene bead, fixed at $\sigma_w = -2\,\mathrm{mC/m^2}, \zeta = -30\,\mathrm{mV}$.

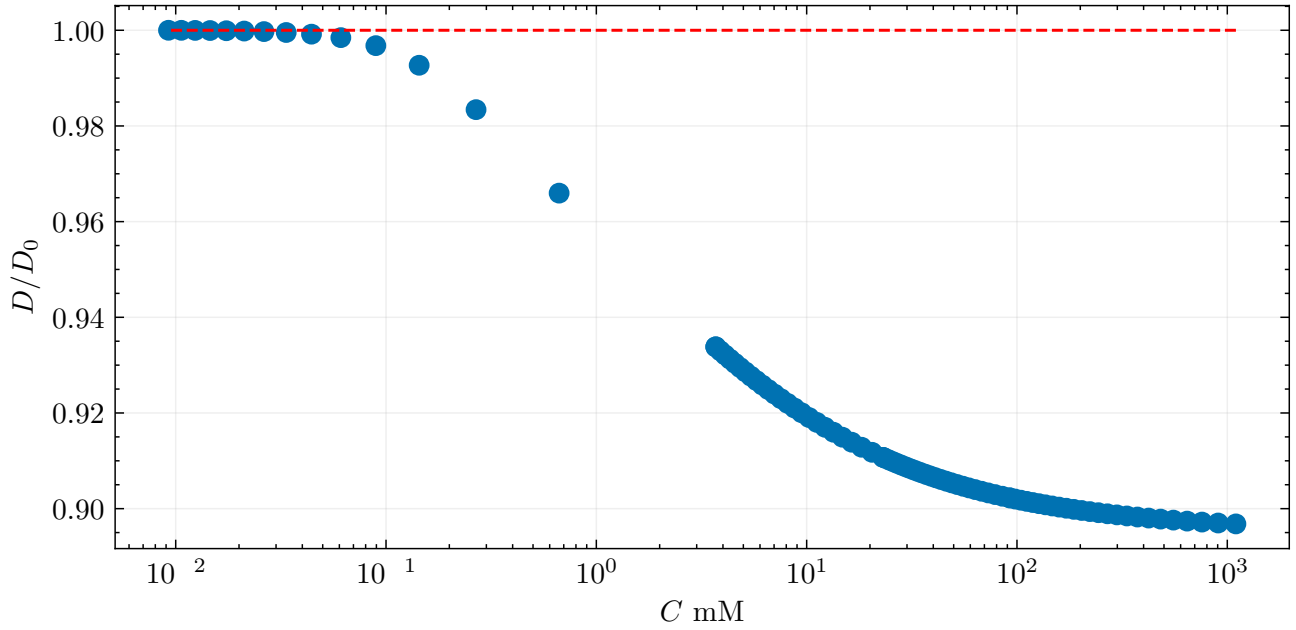


Figure 5.4: Effective diffusion plotted against the solution's debye length. $a = 50\,\mathrm{nm}, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$ and the debye length is varied. The zeta potential of the particle and glass are chosen to resemble glass and a polystyene bead, fixed at $\sigma_w = -2\,\mathrm{mC/m^2}, \zeta = -30\,\mathrm{mV}$.

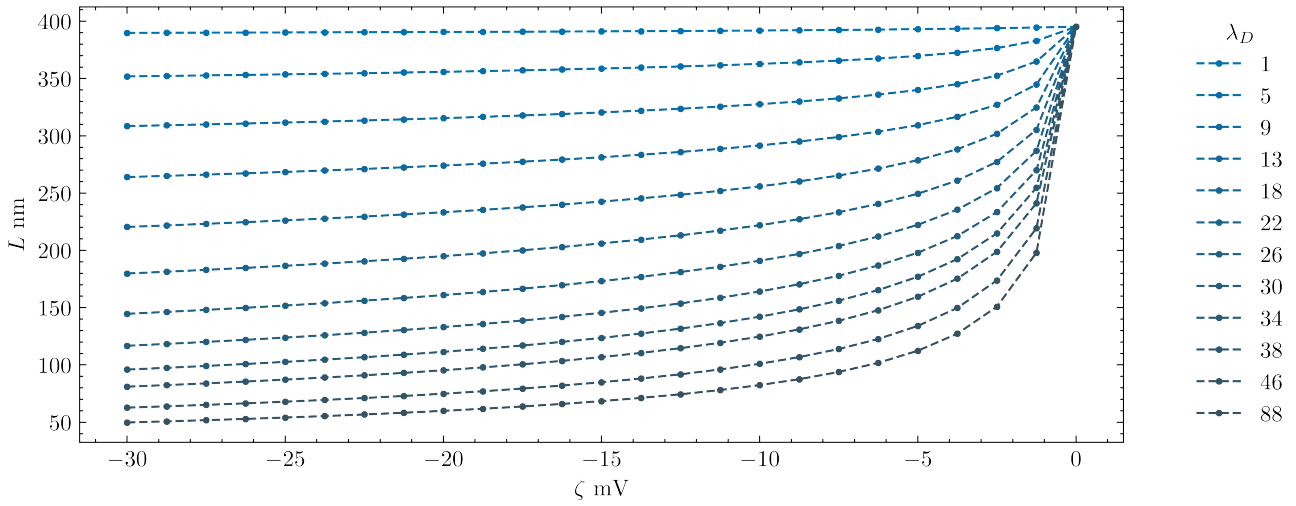


Figure 5.5: Effective confinement plotted against the particle's zeta-potential. $a = 50\,\mathrm{nm}, L = 500\,\mathrm{nm}, T = 300\,\mathrm{K}, \Delta t_i = 0.2\,\mu\mathrm{s}, \tau = 200\,\mu\mathrm{s}, \eta = 1\,\mathrm{mPa\,s}$ and the debye length is varied. The surface charge of the wall are chosen to resemble glass, fixed at $\sigma_w = -2\,\mathrm{mC/m^2}$.

5.3.2 Surface Charge estimation

The particle's exploration demonstrated a clear dependence on a drift due to the electrostatics. This drift is a useful estimator of the particle-wall charge relationship. Using the aforementioned amplitude maximum-likelihood estimator, drift estimator can be extracted into meaningful, physical information.

plot of likelihood of charge or something vs groud truth.

5.4 Conclusion

In conclusion, we have shown that confined simulations can be readily simulated with physical parameters as per literature. we have introduced a new MLE for the amplitude based on ito drift calculations.

Chapter 6

iSCAT z-estimation Simulations

here we discuss the estimation of z-estimation on rayleigh length diffraction. To obtain a 3D track, it is paramount that we have a scope of how accurate the detection of the z-axis is. This section aims to rigourously test and investigate the z-localization method in order to have an accurate representation of the performance of the method.

This chapter first discusses the overview of the fitting method. The theory and statistical tools used are discussed. A general pseudocode is provided. Following this, the model is evaluated on simulated iPSF data using Mie Theory using a published iSCAT iPSF Python library. Speed and performance are taken into account.

6.1 Method Overview

The Axial inference pipeline is described in order of operation on image data.

6.1.1 Image Background Subtraction and Noise Consideration

6.1.2 Radial Integral

Image data will have background misalignment due to iSCAT hardware setup. To optimize for fitting speed and misalignement property, radial scattering profile are assumed as in the rayleigh regime model is symmetric. The radial integration method is described in Alg 1.

Algorithm 1 Discrete Radial Integral

Require: I$\{F, H, W\}$, list of image ROIs
1: θ_0, starting angle $\Delta\theta$, Angular integration distance R, polar integration bound
2: rad_av_arr = []
3: $m = \text{Disk}[R, \theta_0, \theta_0 + \Delta\theta]$
4: **for** $t = 1$ to F **do**
5: $\quad v = I[m]$
6: $\quad \text{rad_av_arr}[t] = \sum_\theta v$
7: **end for**
8: **return** rad_av_arr

6.1.3 Nuisance parameter Estimation

To obtain accurate estimation of the z position relative to nearby frames, nuisance estimation is done to focus on z fitting. To do this, L-BFGS-B Fitting is performed. alongside amplitude fitting. This is described in the algorithm below.

6.1.4 2D Chainlink

To link consequent positions in space, a dynamic programming algorithm is used. The algorithm is descirbed in Alg. 2

Algorithm 2 Dynamic Programming for Best 2D Chain of Points

Require: min_idx = $\{\text{min_idx}[t]\}_{t=0}^{F-1}$, list of candidate positions per frame
min_vals = $\{\text{min_vals}[t]\}_{t=0}^{F-1}$, list of candidate values per frame
$w = (w_x, w_y)$, weighting for distance computation (default $(1, 1)$)
max_step, optional maximum allowed step distance
Ensure: idx_chain, x_chain, y_chain, the indices, positions, and values of the optimal path
Initialize $F \leftarrow |\text{min_idx}|$
Initialize $cost[i] \leftarrow 0$ for all i in frame 0
Initialize $prev \leftarrow []$ ▷ to store predecessor indices
for $t = 1$ to $F - 1$ **do**
    $A_x \leftarrow \text{min_idx}[t-1]$, $B_x \leftarrow \text{min_idx}[t]$
    $A_y \leftarrow \text{min_vals}[t-1]$, $B_y \leftarrow \text{min_vals}[t]$
    Compute pairwise differences:

$$\Delta_x[i,j] = |A_x[i] - B_x[j]|, \quad \Delta_y[i,j] = |A_y[i] - B_y[j]|$$

    Compute weighted Euclidean distances:

$$D[i,j] = \sqrt{(w_x \Delta_x[i,j])^2 + (w_y \Delta_y[i,j])^2}$$

    if max_step is not None **then**
        Set $D[i,j] \leftarrow \infty$ if $D[i,j] > \text{max_step}$
        if no finite $D[i,j]$ exists **then**
            return empty path
        end if
    end if
    Compute cumulative path cost matrix:

$$M[i,j] = cost[i] + D[i,j]$$

    For each node j in frame t, find predecessor $p[j]$ minimizing $M[:,j]$:

$$p[j] = \arg\min_i M[i,j], \quad new_cost[j] = M[p[j], j]$$

    Update $cost \leftarrow new_cost$
    Append p to $prev$
end for
Initialize $j \leftarrow \arg\min_i cost[i]$ ▷ index of minimal total cost in last frame
Initialize idx_chain $\leftarrow [0]^F$, set idx_chain$[F-1] \leftarrow j$
for $t = F - 2$ down to 0 **do**
    idx_chain$[t] \leftarrow prev[t][\text{idx_chain}[t+1]]$
end for
Compute chains:

$$\text{x_chain}[t] = \text{min_idx}[t][\text{idx_chain}[t]], \quad \text{y_chain}[t] = \text{min_vals}[t][\text{idx_chain}[t]]$$

return idx_chain, x_chain, y_chain

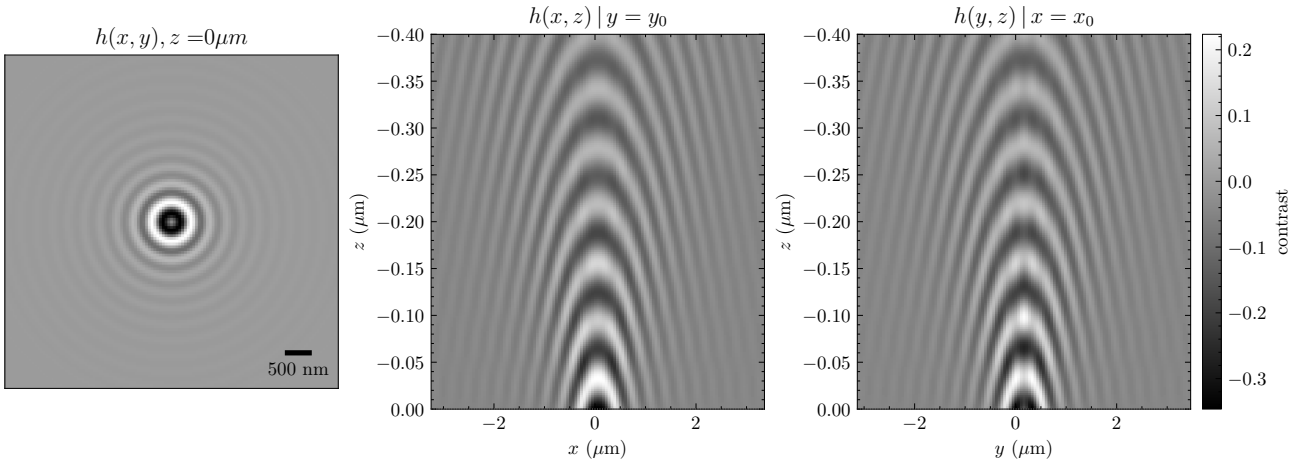


Figure 6.1: Simulated iPSF. Left. Projection of in-focus plane. Middle. x, z projection. Right. y, z projection. Focal scan is over 400 nm. pixel size is 55 nm/pixel. $\lambda = 405$ nm. Assumed particle radius $a = 50$ nm.

6.2 Simulated iPSF localization

To validate the proposed method, one performs simulated axial-estimation. Simulated iSCAT point spread functions were generated using the PySCAT library for $\lambda = 405$ nm. The iPSF was generated for the focal range $z_f \in [-1000, 0]$. The assumed particle is spherical with radius $a = 50$ nm. The iPSF is shown in Fig. 6.1

As mentioned, the localization pipeline utilizes a weighting scheme to account for objective abberation and misalignments to non-modelled regions of the data, such as the central fringe. This is accounted for by using a binary weighting array, $\mathbf{w}$. In the case of these idealized simulations, we will show the effect of changing the weighting scheme as we proceed. This demonstrates a loss of absolute precision, however, maintains the relative precision, which is that of interest in measurements of axial trajectory.

6.2.1 3-parameter Estimation

As a reminder, the model for the iPSF is a multivariate function iPSF : $(z, E_0, \gamma) \to \mathbb{R}^2$. It has been shown that the statistical model of iPSF takes;

$$\begin{aligned} \hat{E_0} &\sim \text{Gamma}(\mu_{E_0}, \sigma_{E_0}) \\ \hat{\gamma} &\sim \text{logNorm}(\mu_\gamma, \sigma_\gamma) \end{aligned} \tag{6.1}$$

The so-called 3-parameter estimation is a sampling of nuisance parameters in the model. The nuisance parameters here are $\hat{E_0}, \hat{\gamma}$, where $\hat{E_0}$ is an ampltiude factor, and γ is an exponential decay factor used to account for spatial decay in signal. The amplitude factor can be analytically solved for using a grid-search sweep along z. The γ factor is fitted to using the L-BFGS-B

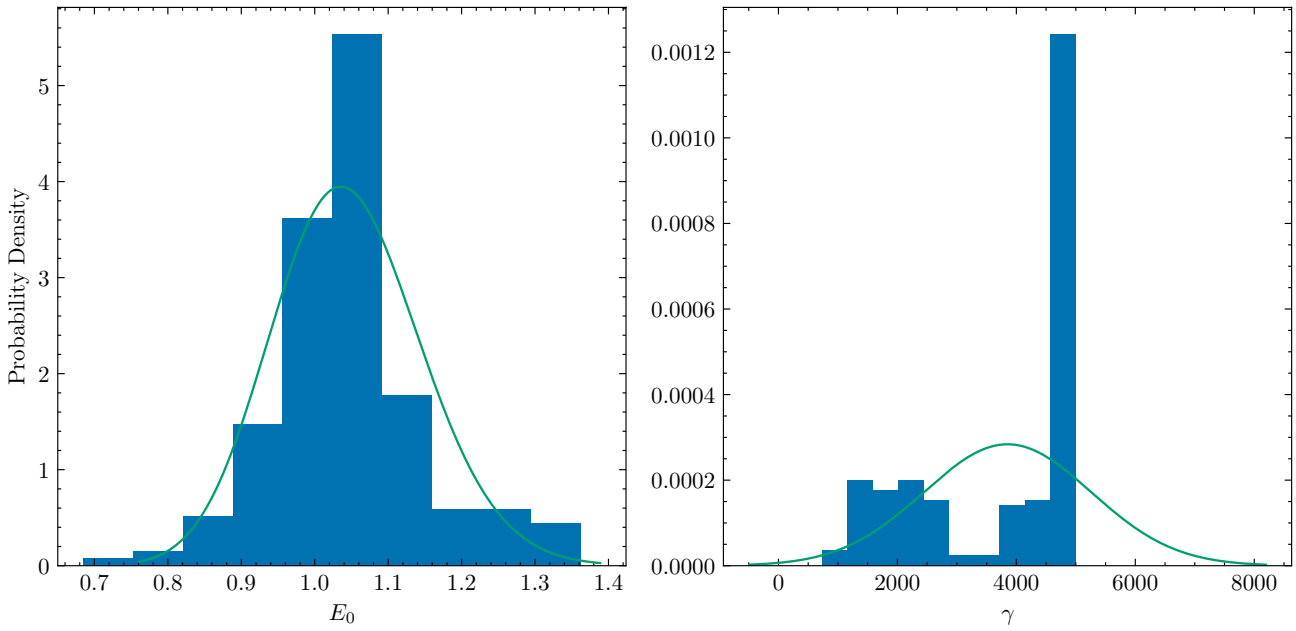


Figure 6.2: Nuisance Parameter estimation for iPSF fits. Left. Ampltiude factor. right. exponential decay factor. The exponential decay is not of utmost importance as it often is very small. it is kept for stability purposes.

minimization algorithm. This nuisance parameter fit is shown in the histograms of these two parameters shown in Fig. 6.2.

6.2.2 Grid-Search Estimation

Once the nuisance parameters are chosen, a gridsearch with a dynamic linking is performed. Each frame is assumed to have a maximal diffusion time. This is in place to optimize for noise and have coherent, physical traces without outliers. The grid search takes the difference between the data and a scan over the model with the nuisance parameters in place. The best loss is chosen. In the case of linking, the minimal loss over the path is chosen. The full algorithm is discussed elsewhere. the residuals of this using no fitting are shown in Fig. 6.3.

6.3 Conclusions

In conclusion, is the method valid for an actual experiment?

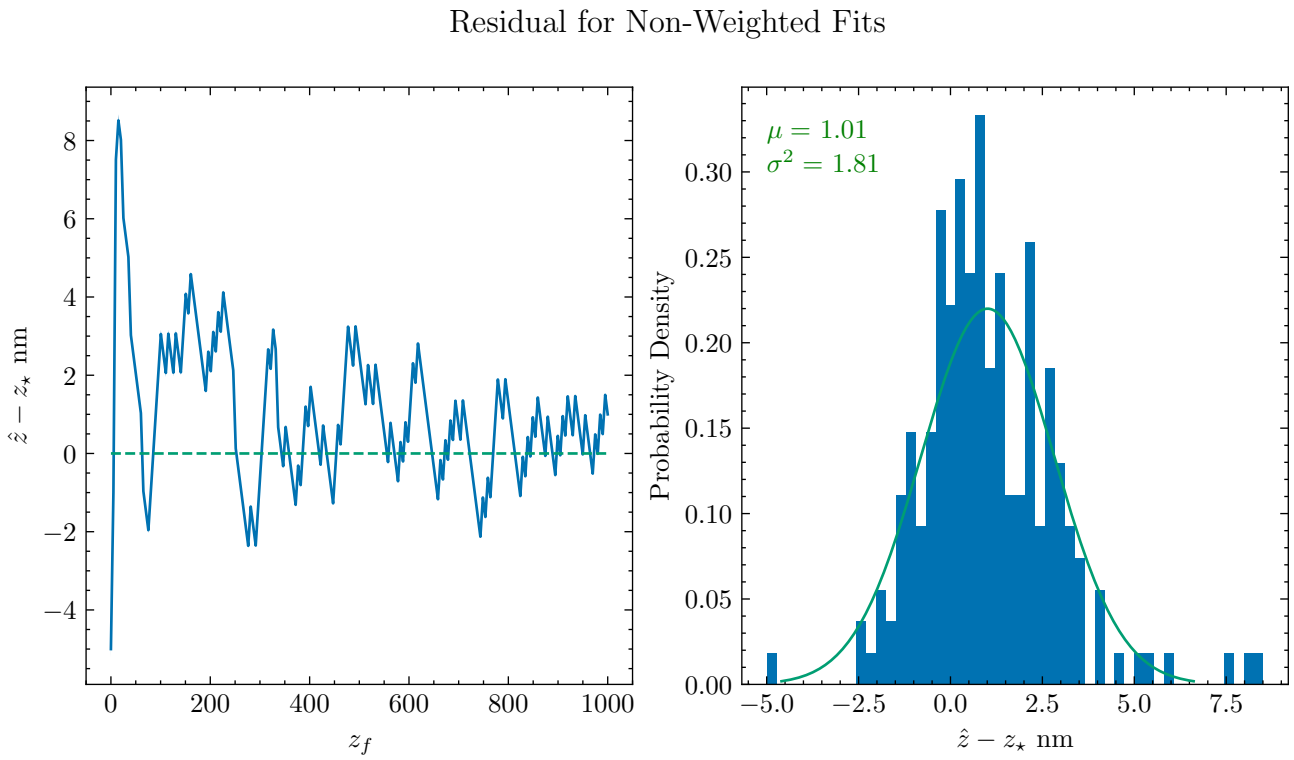


Figure 6.3: Fit resdiuals using no weighting scheme. reportng an absolut bias of 1 nm with variance of 2 nanometers.

Chapter 7

Widefield backscattering iscat for SPT in 3d confinement tracking and analysis

7.1 Questions

What are we trying to ask here

7.2 Reparameterization of Rayleigh scattering of nanoparticles

Cite model used in this thesis

7.3 Fitting model developed in this thesis

Walk through the fitting procedure developed in this thesis. Discuss prior methods, as well as why this method was chosen/altenrative.

use simulated models from mie theory, show exactly how well it performs. Add noise, compare.

7.4 Methods

7.4.1 Sample Details

Describe experimental methods used for passivation, coverslip conditions (basically what i would do for a paper.)

1. Polystyrene Beads measurements in PBS

Sample was purchased from thermofisher (Product ID.). Sample consists of 0.1 um polystyrene beads with 1000nm dfluorescent tag attached. the tag is attached becasue that is what the lab had and has used in prior measurements.

7.4.2 Optical System

Section of optical setup of my iscat system. Discuss jacket of objetive, etc. Exactly details of all the system

7.4.3 Convex-lens induced confinement

describe the covex-lens induced confinement method.

7.5 Results i guess

im not sure at the moment.

7.6 Discussion of Results

Chapter 8

Conclusion

8.1 Conclusion

8.2 Future outlook

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Appendix A

Surface Charge Estimator Derivation

This is a section for the derivation of the surface charge estimator used in the axial diffusion estimation.

A.1 $\hat{B}$ MLE

The Force used in the modelling is given by;

$$V'(z) = B\frac{k_bT}{\lambda_D} \cdot \exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D)) \tag{A.1}$$

The dynamics are governed by the equation, written in the Ito convention,

$$\Delta z = D'(z)\Delta t - \beta D(z)V'(z)\Delta t + \sqrt{2D_z}W_t. \tag{A.2}$$

Re-writing in terms of drift and diffusion coefficients;

$$\Delta z = \left[D_z'(z) - \beta BD(z)\phi(z)\right]\Delta t + \sqrt{2D_z}W_t. \tag{A.3}$$

where $\phi(z) = \frac{k_bT}{\lambda_D}\exp(-H/\lambda_D) \cdot (\exp(-z_n/\lambda_D) - \exp(z_n/\lambda_D))$. This is a Markov-process that has a propagator probability from the Onsager-Machlup path integral representation,

$$P(z_{n+1}|z_n) = \frac{1}{4D_z\Delta t}\exp\left[-\frac{\left(\Delta z_n - D_z'(z) + \beta BD(z)\phi(z)\right)^2}{4D_z\Delta t}\right]. \tag{A.4}$$

To obtain an estimator for B, we can take the log-likelihood and differentiate, maximizing it.

$$
\begin{aligned}
l(B) &= \sum_{n=0}^{\infty} \left[-\frac{\left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2}{4D_z\Delta t} \right] + \text{const} \\
&= -\frac{1}{2}\sum_{n=0}^{\infty} \frac{1}{2D_z\Delta t}\left(\Delta z_n - D_z'(z) + \beta B D(z)\phi(z)\right)^2 + \text{const.}
\end{aligned}
$$

Define $r_n = \Delta z_n - D_z'(z)$ and $w_n = \frac{1}{2D_z}$. Find the maximum of the log-likelihood by taking the derivative of l with respect to B.

$$
l'(B) = -\sum_{n=0}^{\infty} \frac{w_n}{\Delta t}\left(r_n + \beta B D(z)\phi(z)\right)\beta D(z)\phi(z)
$$

We can set this to 0 and solve for $\hat{B}$.

$$
\hat{B} = \frac{\sum_n \frac{\phi(z)}{2D_z}\left[\Delta z - D'(z)\Delta t\right]}{\sum_n \frac{\phi^2(z)\Delta t}{2D_z}} \tag{A.5}
$$

A.2 $\mathbb{V}[\hat{B}]$

The variance of this can be found using the fisher information matrix;

$$
\begin{aligned}
\mathcal{I}(B) &= -l''(B) \\
&= \sum_{n=0}^{\infty} \frac{\Delta t \phi^2(z)}{2D(z)}
\end{aligned}
$$

In the univariate case, the variance is proportional to the inverse of the fisher information, or the denominator of the MLE.

$$
\mathbb{V}[\hat{B}] = \sum_{n=0}^{\infty} \frac{\phi^2(z)\Delta t}{2D(z)} \tag{A.6}
$$

This is an appication of the delta-method as

$$
\begin{aligned}
W_t &\sim \mathcal{N}(0,1) \\
B &= g(W_t).
\end{aligned} \tag{A.7}
$$